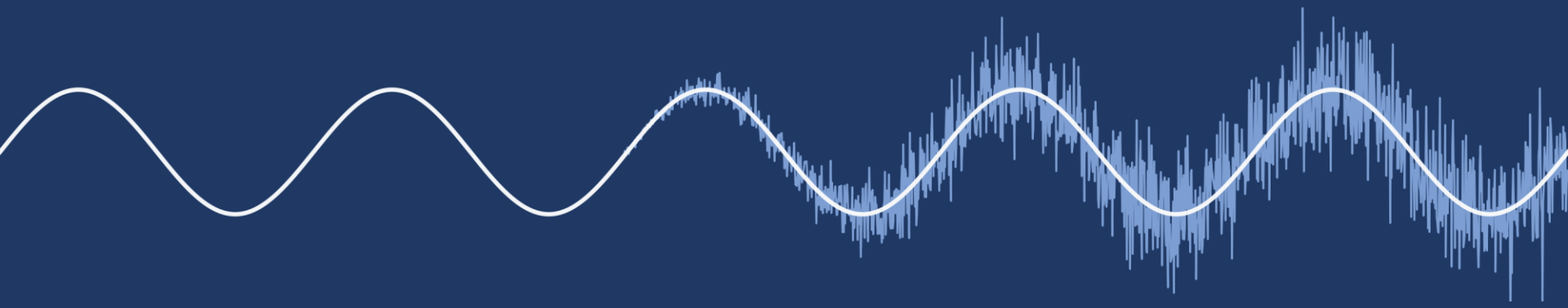
BRAC UNIVERSITY

Department of Electrical and Electronic Engineering, BSRM School of Engineering

Introduction to Communication Engineering

EEE341/ECE341

Lecture 02: Noise in Communication Systems

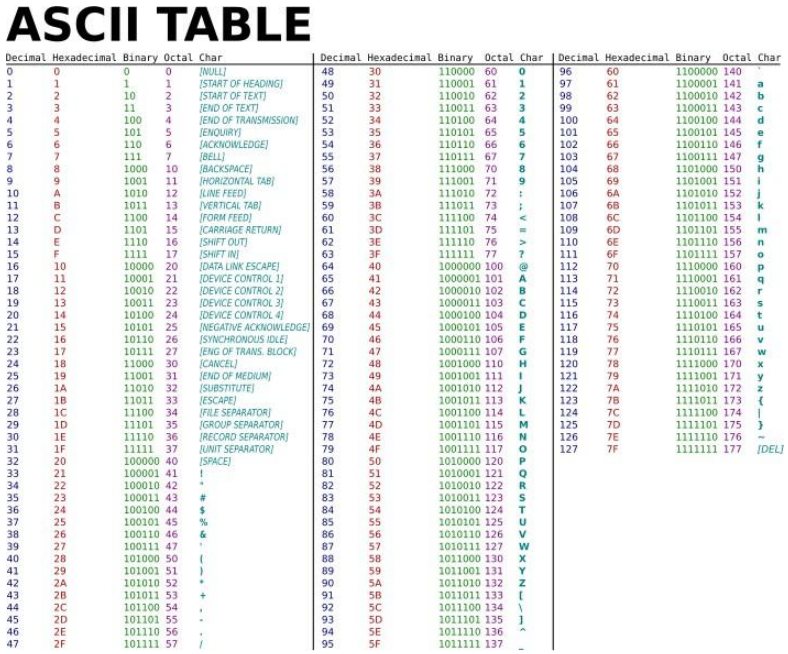


Analog and Digital Messages

Messages are **digital** or **analog**.

Digital messages are **ordered combinations of finite symbols** or codewords.

- **Printed English** consists of 26 letters, 10 numbers, a space, and several punctuation marks. A text document written in English is a digital message constructed from the ASCII keyboard of **128 symbols (2^7, i.e. 7-bit codewords)**.
- A **Morse-coded telegraph message** is constructed from a set of only **two symbols, dash and dot**. It is therefore a **binary** message.
- A digital message constructed with **M symbols** is called an **M-ary message**.



ASCII TABLE

Decimal	Hexadecimal	Binary	Octal	Char
0	0	0	0	[NULL]
1	1	1	1	[START OF HEADING]
2	2	10	2	[START OF TEXT]
3	3	11	3	[END OF TEXT]
4	4	100	4	[END OF TRANSMISSION]
5	5	101	5	[ENQUIRY]
6	6	110	6	[ACKNOWLEDGE]
7	7	111	7	[BELL]
8	8	1000	10	[BACKSPACE]
9	9	1001	11	[HORIZONTAL TAB]
10	A	1010	12	[LINE FEED]
11	B	1011	13	[VERTICAL TAB]
12	C	1100	14	[FORM FEED]
13	D	1101	15	[CARRIAGE RETURN]
14	E	1110	16	[SHIFT OUT]
15	F	1111	17	[SHIFT IN]
16	10	10000	20	[DATA LINK ESCAPE]
17	11	10001	21	[DEVICE CONTROL 1]
18	12	10010	22	[DEVICE CONTROL 2]
19	13	10011	23	[DEVICE CONTROL 3]
20	14	10100	24	[DEVICE CONTROL 4]
21	15	10101	25	[NEGATIVE ACKNOWLEDGE]
22	16	10110	26	[SYNCHRONOUS IDLE]
23	17	10111	27	[ENG OF TRANS. BLOCK]
24	18	11000	30	[CANCEL]
25	19	11001	31	[END OF MEDIUM]
26	1A	11010	32	[SUBSTITUTE]
27	1B	11011	33	[ESCAPE]
28	1C	11100	34	[FILE SEPARATOR]
29	1D	11101	35	[GROUP SEPARATOR]
30	1E	11110	36	[RECORD SEPARATOR]
31	1F	11111	37	[UNIT SEPARATOR]
32	20	100000	40	[SPACE]
33	21	100001	41	!
34	22	100010	42	"
35	23	100011	43	#
36	24	100100	44	$
37	25	100101	45	%
38	26	100110	46	&
39	27	100111	47	'
40	28	101000	50	(
41	29	101001	51	)
42	2A	101010	52	*
43	2B	101011	53	+
44	2C	101100	54	,
45	2D	101101	55	-
46	2E	101110	56	.
47	2F	101111	57	/
48	30	110000	60	0
49	31	110001	61	1
50	32	110010	62	2
51	33	110011	63	3
52	34	110100	64	4
53	35	110101	65	5
54	36	110110	66	6
55	37	110111	67	7
56	38	111000	70	8
57	39	111001	71	9
58	3A	111010	72	:
59	3B	111011	73	;
60	3C	111100	74	<
61	3D	111101	75	=
62	3E	111110	76	>
63	3F	111111	77	?
64	40	1000000	100	@
65	41	1000001	101	A
66	42	1000010	102	B
67	43	1000011	103	C
68	44	1000100	104	D
69	45	1000101	105	E
70	46	1000110	106	F
71	47	1000111	107	G
72	48	1001000	110	H
73	49	1001001	111	I
74	4A	1001010	112	J
75	4B	1001011	113	K
76	4C	1001100	114	L
77	4D	1001101	115	M
78	4E	1001110	116	N
79	4F	1001111	117	O
80	50	1010000	120	P
81	51	1010001	121	Q
82	52	1010010	122	R
83	53	1010011	123	S
84	54	1010100	124	T
85	55	1010101	125	U
86	56	1010110	126	V
87	57	1010111	127	W
88	58	1011000	130	X
89	59	1011001	131	Y
90	5A	1011010	132	Z
91	5B	1011011	133	[
92	5C	1011100	134	\
93	5D	1011101	135	]
94	5E	1011110	136	^
95	5F	1011111	137	_
96	60	1100000	140	`
97	61	1100001	141	a
98	62	1100010	142	b
99	63	1100011	143	c
100	64	1100100	144	d
101	65	1100101	145	e
102	66	1100110	146	f
103	67	1100111	147	g
104	68	1101000	150	h
105	69	1101001	151	i
106	6A	1101010	152	j
107	6B	1101011	153	k
108	6C	1101100	154	l
109	6D	1101101	155	m
110	6E	1101110	156	n
111	6F	1101111	157	o
112	70	1110000	160	p
113	71	1110001	161	q
114	72	1110010	162	r
115	73	1110011	163	s
116	74	1110100	164	t
117	75	1110101	165	u
118	76	1110110	166	v
119	77	1110111	167	w
120	78	1111000	170	x
121	79	1111001	171	y
122	7A	1111010	172	z
123	7B	1111011	173	{
124	7C	1111100	174	|
125	7D	1111101	175	}
126	7E	1111110	176	~
127	7F	1111111	177	[DEL]

Fig. ASCII table [3]

Analog Messages

Analog messages are characterized by data whose **values vary over a continuous range** and are defined for **a continuous range of time**.

- A particular **speech waveform** has **amplitudes that vary over a continuous range**.
- Over a given time interval, an infinite number of possible different speech waveforms exist, in contrast to only a finite number of possible digital messages.

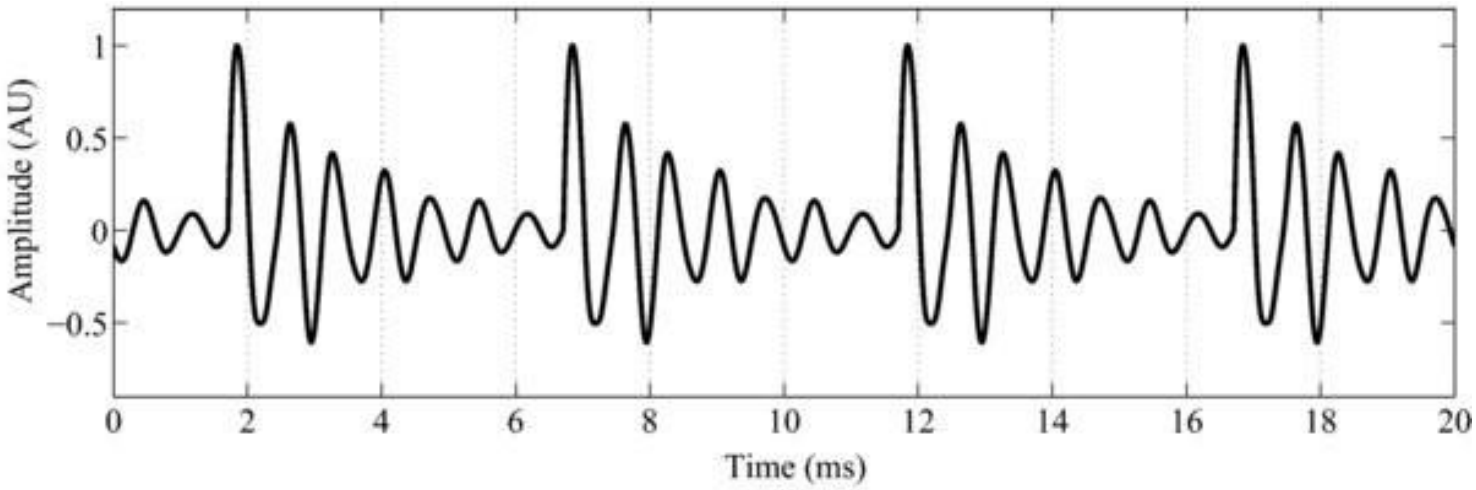


Fig. A speech waveform [4]

Why Are Digital Technologies Better?

The answer has to do with both **economics** and **quality**.

Economics

The case for economics is made by noting the **ease of adopting** versatile, powerful, and inexpensive high-speed digital microprocessors.

Quality

A prominent feature of digital communications is the **enhanced immunity of digital signals to noise and interference**.

Digital messages are transmitted as a **finite set of electrical waveforms**.

A digital message is generated from a **finite alphabet**, while each character in the alphabet can be represented by one waveform or a sequential combination of such waveforms.

Digital Messages as Electrical Waveforms

For example, in sending messages via **Morse code**:

- a **dash** can be transmitted by an electrical pulse of amplitude **A/2**
- a **dot** can be transmitted by a pulse of negative amplitude **−A/2** (Fig. 1.3a)

In an **M-ary case**, **M distinct electrical pulses** (or waveforms) are used; each of the M pulses represents one of the M possible symbols.

Once transmitted, the receiver must extract the message from a **distorted and noisy signal** at the channel output.

Message extraction is often easier from digital signals than from analog signals because the digital decision must belong to the **finite-sized alphabet**.

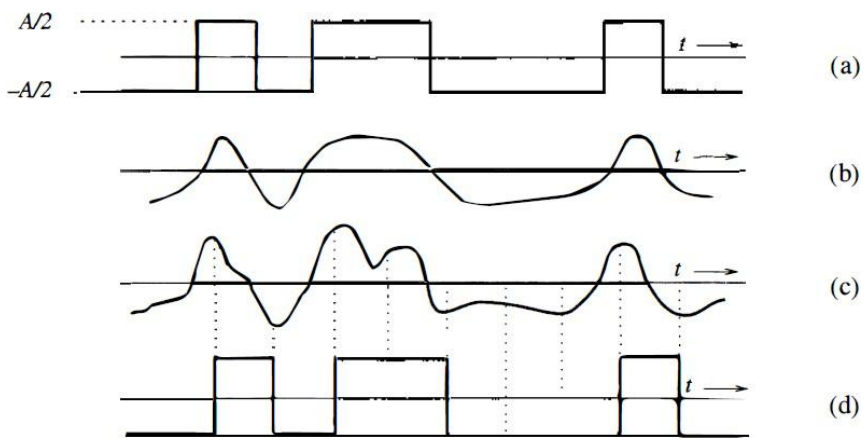


Figure 1.3. (a) Transmitted signal. (b) Received distorted signal (without noise). (c) Received distorted signal (with noise). (d) Regenerated signal (delayed).

Noise Immunity of Digital Signals

Consider a binary case: two symbols are encoded as rectangular pulses of amplitudes **A/2** and **−A/2**.

- The only decision at the receiver is to select between **two possible pulses** received; the fine details of the pulse shape are not an issue.
- A **finite alphabet** leads to **noise and interference immunity**.
- The receiver's decision can be made with reasonable certainty even if the pulses have suffered **modest distortion and noise** (Fig. 1.3).
- The digital message in Fig. 1.3a is distorted by the channel, as shown in Fig. 1.3b. Yet, if the distortion is not too large, we can recover the data without error.
- Figure 1.3c shows the same data with channel distortion and noise. Here again, the data can be **recovered correctly** as long as the distortion and the noise are within limits.

We need to make only a simple binary decision: **Is the received pulse positive or negative?**

Digital vs. Analog Systems

Digital systems

- The digital decision must belong to a **finite-sized alphabet**.
- Only a **simple binary decision** is needed; the fine details of the pulse shape are not an issue.
- Data can be recovered correctly as long as the distortion and the noise are **within limits**.

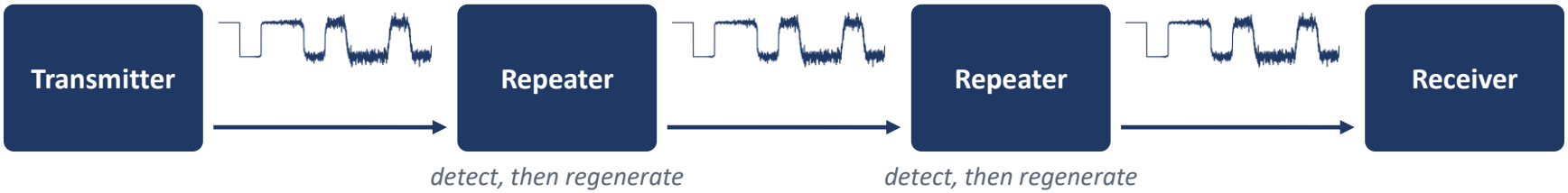
Analog systems

- The **waveform shape itself** in an analog message carries the needed information.
- Even a **slight distortion or interference** in the waveform will show up in the received signal.

A digital communication system is **more rugged** than an analog communication system in the sense that it can better withstand noise and distortion (as long as they are within a limit).

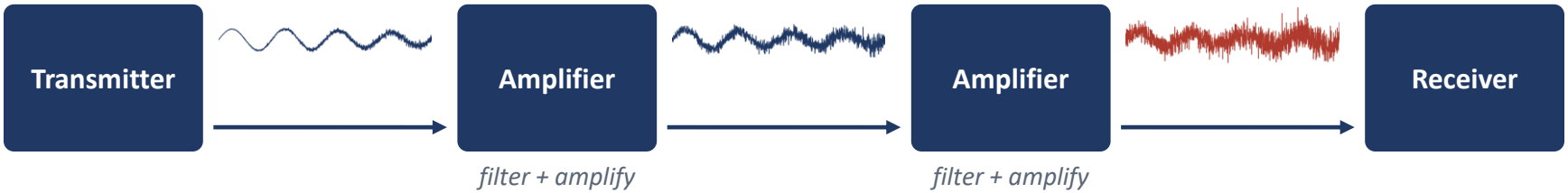
Regenerative Repeaters in Digital Systems

One main reason for the superior quality of digital systems over analog ones is the **viability of regenerative repeaters and network nodes**.



- Repeater stations are **placed along the communication path** at distances short enough to ensure that noise and distortion remain within a limit. This allows pulse detection with high accuracy.
- At each repeater station, or network node, the incoming pulses are detected such that **new, "clean" pulses are retransmitted** to the next repeater station or node.
- This process **prevents the accumulation of noise and distortion** along the path by cleaning the pulses at regular repeater intervals.
- We can thus transmit messages **over longer distances with greater accuracy**.

Repeaters in Analog Systems



- Signals and noise **within the same bandwidth cannot be separated**.
- Repeaters in analog systems are basically **filters plus amplifiers** and are **not "regenerative."**
- Thus, it is impossible to avoid the **in-band accumulation** of noise and distortion along the path.
- The signal is **attenuated continuously** over the transmission path: with increasing distance the signal becomes weaker, whereas the distortion and the noise accumulate more.
- Ultimately, the signal, overwhelmed by distortion and noise, is **buried beyond recognition**.

Amplification is of little help since it enhances both the signal and the noise equally. The distance over which an analog message can be successfully received is limited by the first transmitter's power.

Analog Communication Today

In the past	Despite these limitations, analog communication was used **widely and successfully** for short- to medium-range communications.
Nowadays	Because of the advent of **optical fiber communications** and the dramatic cost reduction achieved in the fabrication of high-speed digital circuitry and digital storage devices, **almost all new communication systems** being installed are digital.
Still in use	Some old analog communication facilities are still in use, including those for **AM and FM radio broadcasting**.

Channel Effect, SNR, and Capacity

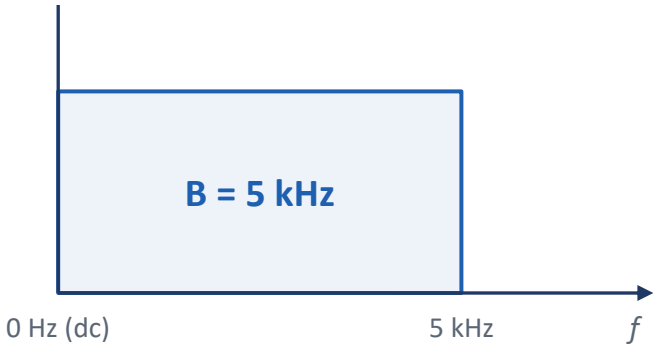
In **designing communication systems**, it is important to understand and analyze important factors such as:

1. the channel characteristics
2. the signal characteristics
3. the relative noise strength
4. the maximum number of bits that can be sent over a channel per second
5. and, ultimately, the signal quality

Signal Bandwidth

The **bandwidth of a channel** is the **range of frequencies that it can transmit with reasonable fidelity**.

- If a channel can transmit with reasonable fidelity a signal whose frequency components vary from 0 Hz (dc) up to a maximum of 5000 Hz (5 kHz), the **channel bandwidth B is 5 kHz**.
- Each signal also has a bandwidth that measures the maximum range of its frequency components.
- The **faster a signal changes**, the higher its maximum frequency is, and the larger its bandwidth is.
- A signal can be successfully sent over a channel if the **channel bandwidth exceeds the signal bandwidth**.



Signals rich in content that changes quickly (**battle scenes in a video**) have larger bandwidth than signals that are dull and vary slowly (**a video of sleeping animals**).

The number of pulses per second that can be transmitted over a channel is **directly proportional to its bandwidth B**: if bandwidth B carries N pulses per second, then KN pulses per second need bandwidth KB.

Signal Power: A Dual Role

The **signal power P_s** plays a **dual role** in information transmission.

First: quality of transmission

- Increasing P_s strengthens the signal pulse and diminishes the effect of channel noise and interference.
- The quality of either analog or digital communication systems varies with the **signal-to-noise ratio (SNR)**.
- A certain minimum SNR at the receiver is necessary for successful communication.
- A larger P_s maintains the minimum SNR over a longer distance, **enabling successful communication over a longer span**.

Second: exchangeable with bandwidth

- From the information theory point of view, the channel bandwidth B and the signal power P_s are, to some extent, **exchangeable**.
- To maintain a given rate and accuracy of information transmission, we can **trade P_s for B**, and vice versa:
  - one may use less B if one is willing to increase P_s
  - or one may reduce P_s if one is given a bigger B

Two Primary Resources: Bandwidth and Power

The two primary communication resources are **bandwidth** and **transmit power**. In a given communication channel, one resource may be more valuable than the other, and the communication scheme should be designed accordingly.

A typical telephone channel

B = 3 kHz

Limited bandwidth, but the power is less restrictive.

Space vehicles

Huge B

Huge bandwidth is available, but the power is severely limited.

Hence, the communication solutions in the two cases are **radically different**.

Channel Capacity and Data Rate

Channel bandwidth limits the bandwidth of signals that can successfully pass through, whereas **signal SNR at the receiver** determines the recoverability of the transmitted signals.

- **Higher SNR** means that the transmitted signal pulse can use more signal levels, thereby carrying more bits with each pulse transmission.
- **Higher bandwidth B** also means that one can transmit more pulses (faster variation) over the channel.
- Hence, SNR and bandwidth B can both affect the underlying channel "throughput."

The AWGN channel

One of the most commonly encountered channels is the **additive white Gaussian noise (AWGN)** channel.

- Assumes no channel distortions except for the additive white Gaussian noise and its finite bandwidth B
- Captures application cases with distortionless channels
- Provides a **performance upper bound** for more general distortive channels

The peak throughput that can be reliably carried by a channel is defined as the **channel capacity**.

Shannon's Capacity Equation

The band-limited AWGN channel capacity was dramatically highlighted by **Shannon's equation**:

$$C = B\log_2(1+\mathrm{SNR}) \ \ \text{bit/s} \tag{1.1}$$

- C is the **upper bound of the rate of information transmission** per second: the maximum number of bits that can be transmitted per second with a probability of error arbitrarily close to zero.
- The capacity **only points out this possibility**; it does not specify how it is to be realized.
- It is **impossible to transmit at a rate higher than this** without incurring errors. Practical systems operate at rates below the Shannon rate.
- B and SNR represent the **ultimate limitation on the rate of communication**; the equation also demonstrates the possibility of resource exchange between these two parameters.
- The trade is **not one-to-one**: doubling the speaker's volume allows the speaker to talk a little faster, but not twice as fast. Doubling the SNR cannot always compensate for the loss of B by 50%.

Example: Channel Capacity

PROBLEM

The bandwidth of a communication channel is 10 MHz. If the received signal power is −60 dB and the noise power is −120 dB/Hz, what will be the channel's capacity?

Channel bandwidth	Received signal power	Noise power density	Required
B = 10 MHz	**S = −60 dB**	**N_0 = −120 dB/Hz**	**C = ?**

- The noise is given per hertz (a noise power spectral density N_0), so the total noise power in the channel is $N = N_0 \times B$.
- Both powers are taken relative to the same reference (for example dBW and dBW/Hz), so the reference cancels in the SNR.

Solution: Channel Capacity

1 **Total noise power in the channel bandwidth**

$N = N_0 \times B \quad \Rightarrow \quad N\,[\text{dB}] = N_0\,[\text{dB/Hz}] + 10 \log_{10} B$

$N = -120 + 10 \log_{10}(10^7) = -120 + 70 = \mathbf{-50\ dB}$

2 **Signal-to-noise ratio**

$\text{SNR}\,[\text{dB}] = S - N = -60 - (-50) = \mathbf{-10\ dB}$

$\text{SNR} = 10^{-10/10} = \mathbf{0.1}$

3 **Shannon's equation**

$C = B \log_2(1 + \text{SNR}) = 10^7 \times \log_2(1.1) = 10^7 \times 0.1375$

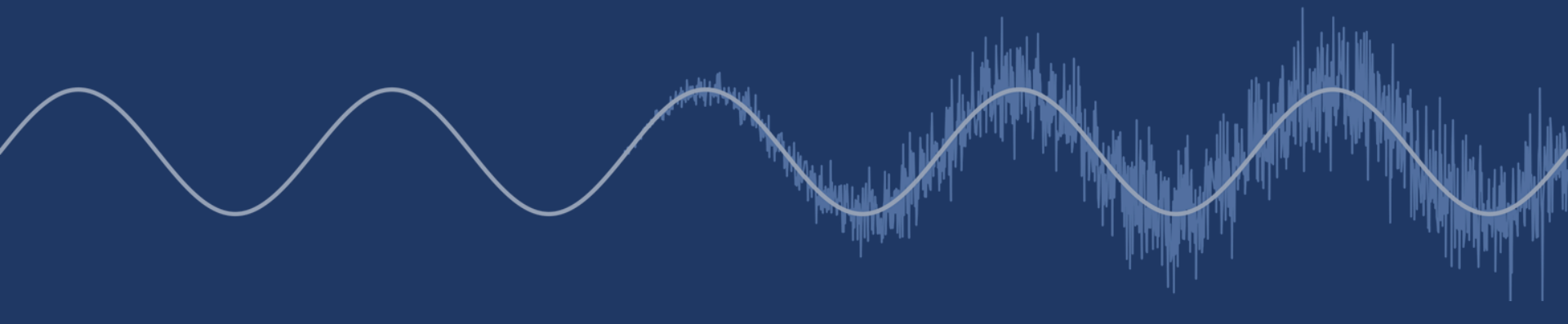
$$\mathbf{C \approx 1.375 \times 10^6\ \text{bit/s} \approx 1.38\ \text{Mbit/s}}$$

The signal is 10 dB below the noise (SNR < 1), yet the capacity is not zero.

PART 2

Noise in Communication Systems

Sources and types of noise | Noise figure | Noise temperature | dB units



Noise

Noise

- **Any unwanted signal**, whether audible or not
- Noise gets added to the signal and degrades the quality of signal

Noise and interference

Although they play a somewhat similar role in electrical systems, they are dissimilar in nature in one important aspect.

Noise

is usually composed of **randomly occurring voltages**, which are unrelated in phase or frequency and may sometimes be of a very peaky nature.

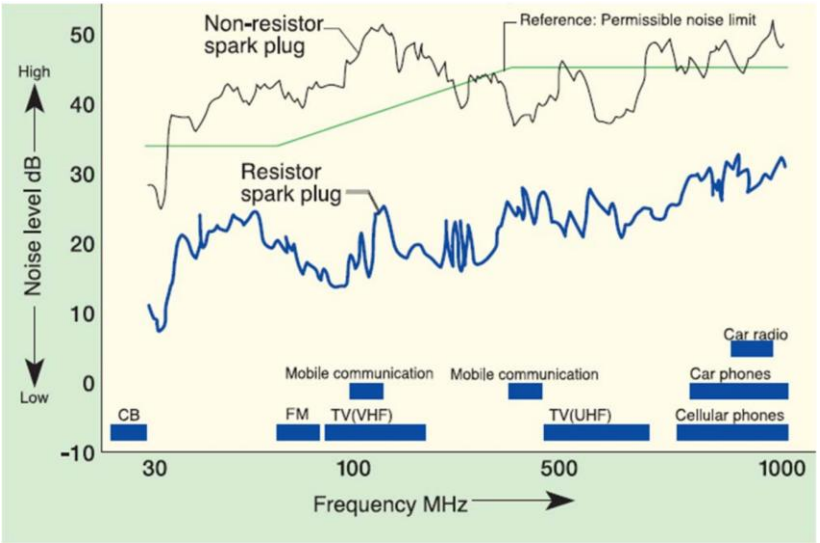
Interference

is usually **more structured** than noise since it arises as unwanted coupling from just a few signals (e.g., from other users) in the network.

Sources of Noise

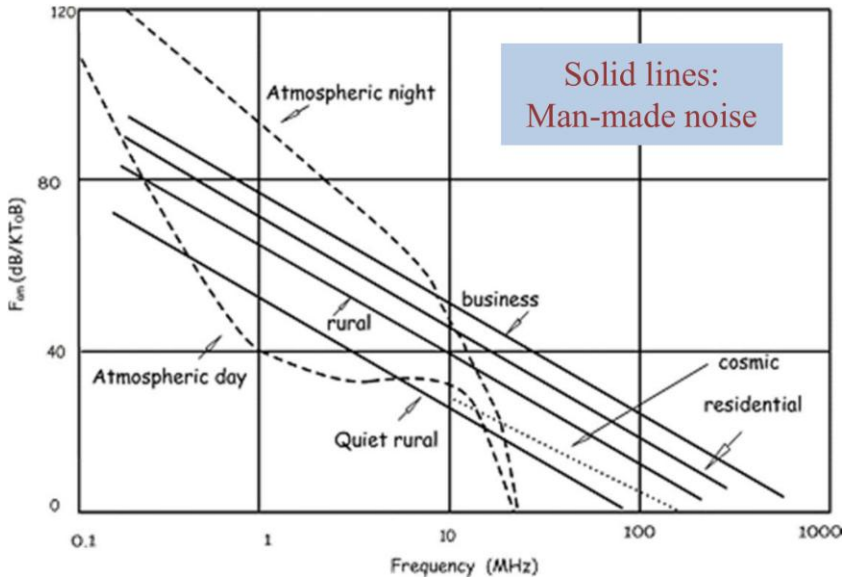
Artificial or man-made sources

- Commutator motors
- Spark plugs of vehicles
- Faulty switches
- Fluorescent lights
- Electric shavers
- Power lines



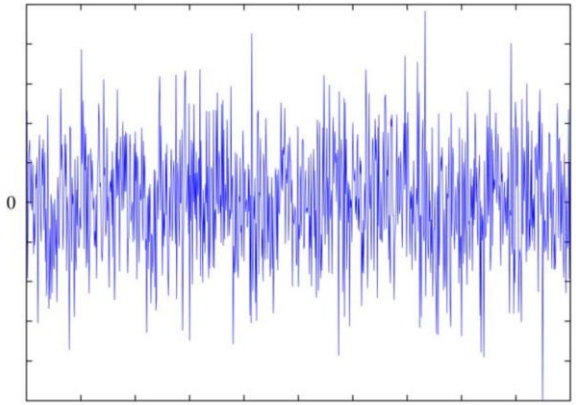
Natural sources

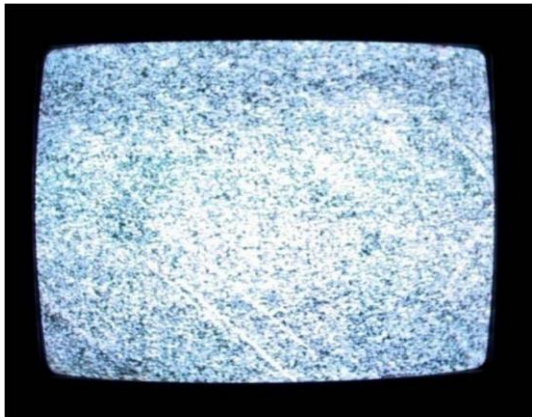
- Cosmic (galactic) radiation
- Atmospheric (e.g., lightning discharge, rain attenuation)
- Intrinsic circuit noise
- etc.



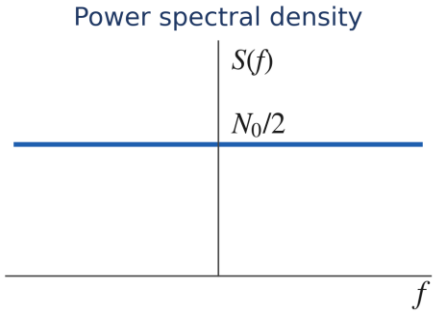
White Noise

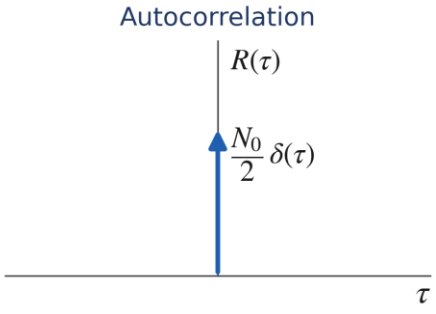
- **Both thermal and shot noise are characterized as white noise**
- White noise means it contains noise of all frequencies with a flat PSD





$$S(f) = \frac{N_0}{2}$$





$$R(\tau) = \frac{N_0}{2}\delta(\tau)$$

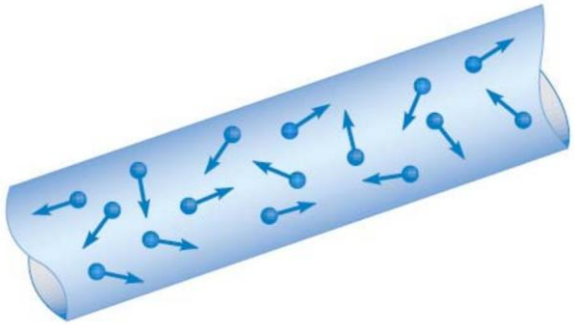
Types of Noise: 1. Thermal Noise

- It is produced by **the random motion of "free" electrons** in a conductor.
- Any substance with temperature above zero kelvin (absolute zero) contains some electrons that are free to move about in that substance. The amount of energy contained by these electrons increases as the temperature increases, and an increase in energy means an increase in the average speeds of the free electrons.
- However, **moving electrons constitute an electric current** as electrons randomly collide with lattice atoms. Since the currents increase with temperature, the **noise power likewise increases with temperature**.

Properties

1. Thermal noise is **present in any conductor**.
2. The **only predictable property** of thermal noise is its **average power**.
3. In the case of thermal noise, the power is spread uniformly up to very high frequency (about a 10% drop at 2 THz).

⇒ **Thermal noise characteristic is white.**



Thermal Noise: Voltage and Power

The mean-square **thermal noise voltage** at the terminals of an open-circuit resistor of value R_s is:

$$\overline{V_n^2} = 4kTBR_s$$

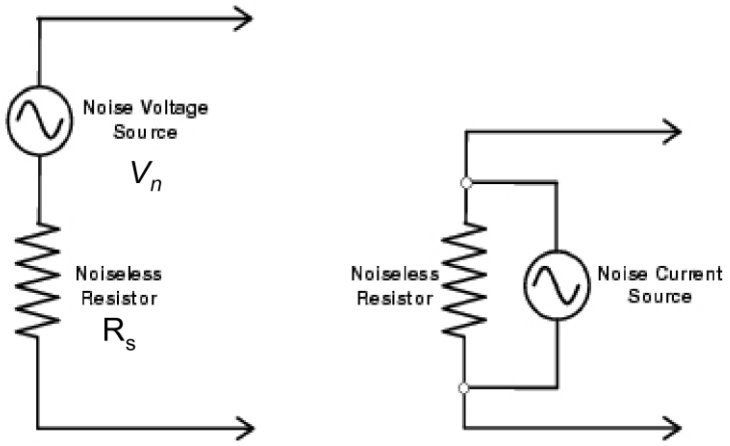
- $\mathbf{V_n^2}$ (with overbar) = mean-square value of thermal voltage
- **k** = Boltzmann's constant = 1.38×10^{-23} J/K
- **T** = absolute temperature of the resistor in kelvin
- **B** = equivalent noise bandwidth (system bandwidth) in Hz
- $\mathbf{R_s}$ = resistance of the conductor

Thermal noise power (maximum):

$$P_n = \frac{\overline{V_n^2}}{(2R_s)^2} \times R_s = \frac{4kTBR_s^2}{4R_s^2} = kTB$$

$$P_n\,[\text{dBm}] = -174 + 10\log_{10} B$$

Unit: dBm, with T = 300 K and B in Hz



Types of Noise: 2. Shot Noise

- Generated in active electronic components due to **discrete and random emission of electrons**
- Shot noise normally occurs when there is a **potential barrier** (voltage differential). A PN-junction diode is an example that has a potential barrier. When the electrons and holes cross the barrier, shot noise is produced.
- A diode, a transistor, and a vacuum tube will all produce shot noise.
- A resistor normally does not produce shot noise since there is no potential barrier built within a resistor.
- If the active device provides amplification, the **noise also gets amplified along with the signal**.
- **Shot noise characteristic is white**

$$\overline{i_n^2} = 2 e I_a B$$

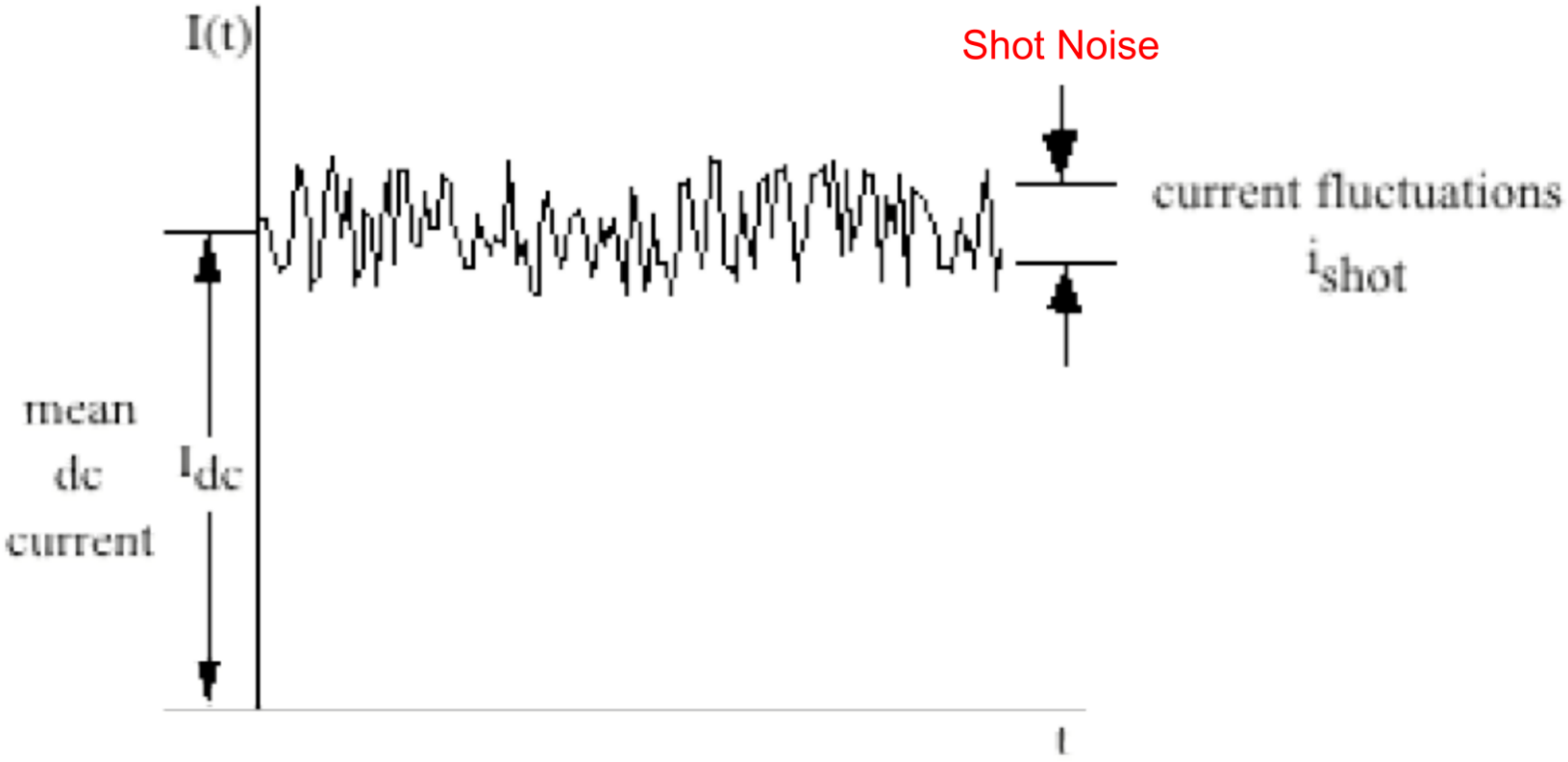
$\mathbf{i_n}$ = RMS value of shot noise current

e = charge of an electron

$\mathbf{I_a}$ = average current

B = bandwidth of the system

Shot Noise: Current Fluctuations



Shot noise: random current fluctuations i_{shot} superimposed on the mean dc current I_{dc}

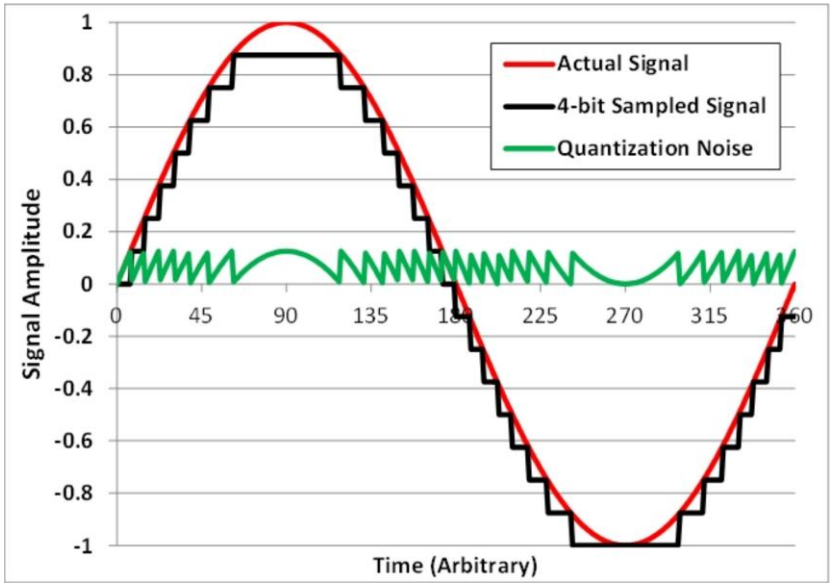
Types of Noise

3. Impulse noise

- Impulse noise can occur from **switching transients in electromechanical switching offices** or from rotary dial telephones
- Step-by-step switching is the most frequent source
- Impulse noise is usually **measured in terms of number of pulses per second**

4. Quantization noise

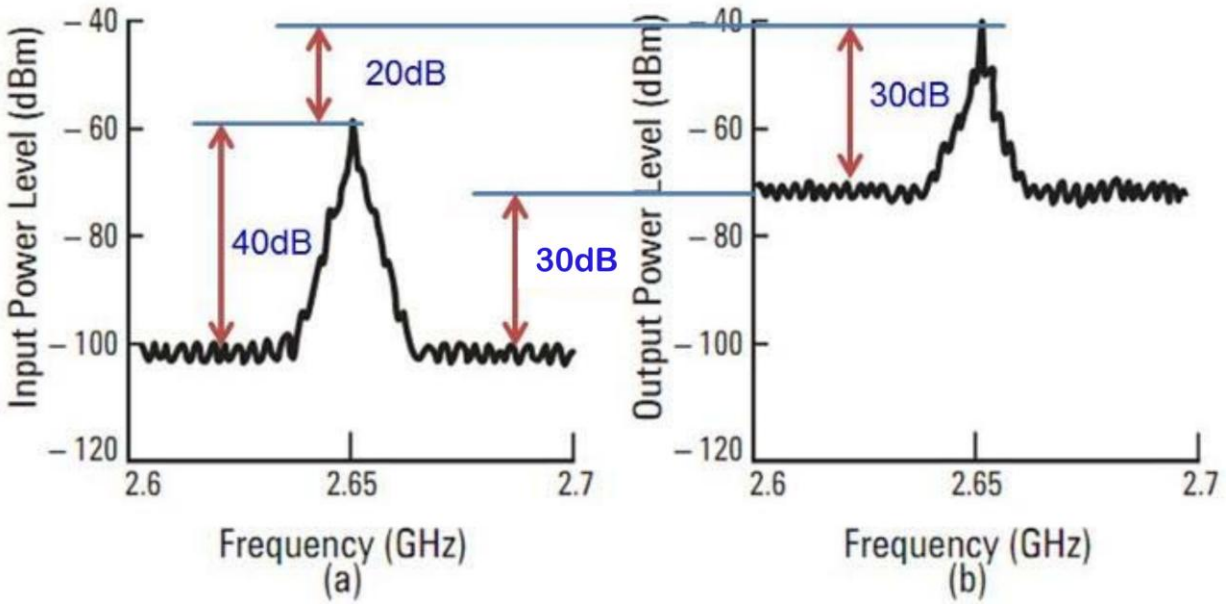
- Quantization **noise arises during the digitization process** as the sampled values are different from the quantized values



Noise Figure (NF)

Noise figure F relates the **SNR at the input** of a network (or device) to the **SNR at the output** of the network (or device).

$$F = \frac{(\mathrm{SNR})_{in}}{(\mathrm{SNR})_{out}} = \frac{P_{in}/N_{in}}{P_{out}/N_{out}}$$



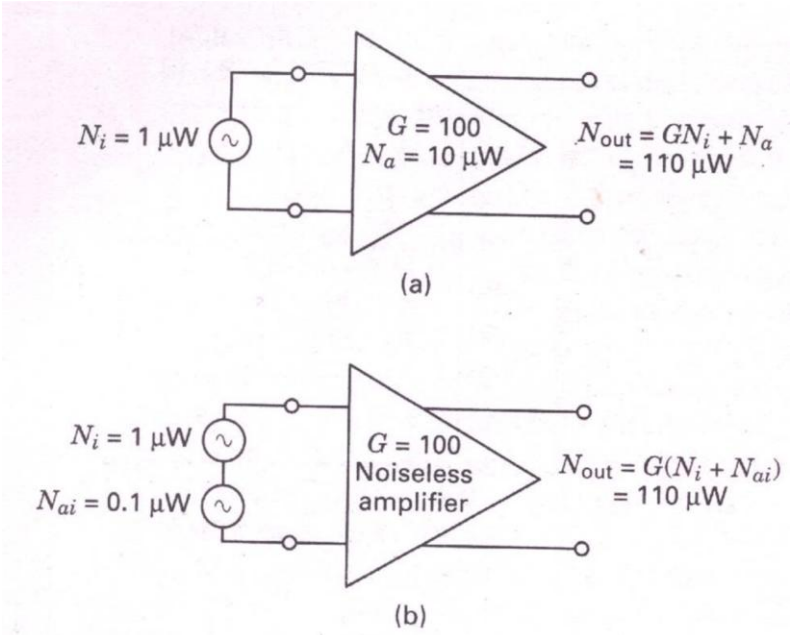
Input SNR = 40 dB

Output SNR = 30 dB

NF = 10 dB

Noise Figure of an Amplifier

$$F = \frac{(\mathrm{SNR})_{in}}{(\mathrm{SNR})_{out}} = \frac{N_o}{G N_i}$$



S_i = signal power at the amplifier input

N_i = noise power at the amplifier input

N_o = noise power at the amplifier output

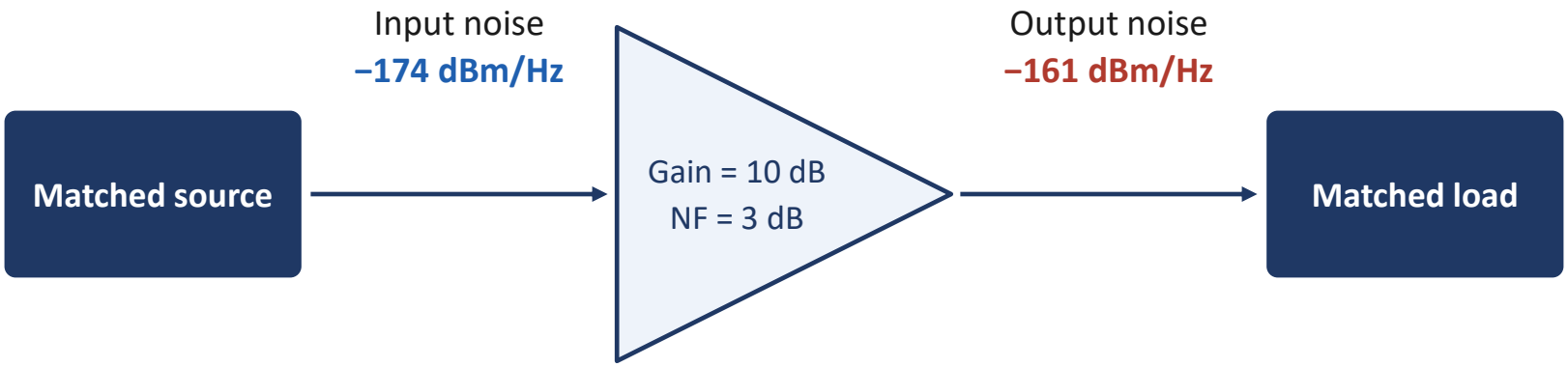
= $G N_i + N_a = G (N_i + N_{ai})$

N_{ai} = amplifier noise referred to the input

G = amplifier gain

$$F = \frac{\frac{S_i}{N_i}}{\frac{G S_i}{G(N_i + N_{ai})}} = 1 + \frac{N_{ai}}{N_i} = 1 + \frac{N_a}{G N_i}$$

Noise Figure: Example



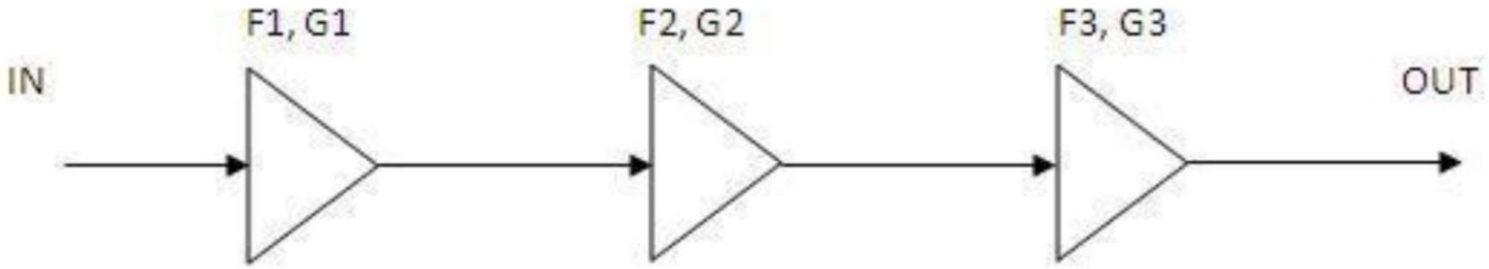
Output noise = input noise + gain + NF

= −174 dBm/Hz + 10 dB + 3 dB = **−161 dBm/Hz**

−174 dBm/Hz is the thermal noise power per hertz (kT) delivered by a matched source at room temperature.

Noise Figure: Cascaded Stages

For an **n-stage cascaded system**:



$$F = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \cdots + \frac{F_n - 1}{G_1 G_2 \cdots G_{n-1}}$$

Cascaded noise figure

NF (dB) = 10 $\log_{10}$(F)

Cascaded gain

$G = G_1 \times G_2 \times G_3$

$G\ (dB) = G_1\ (dB) + G_2\ (dB) + G_3\ (dB)$

Noise Temperature (1)

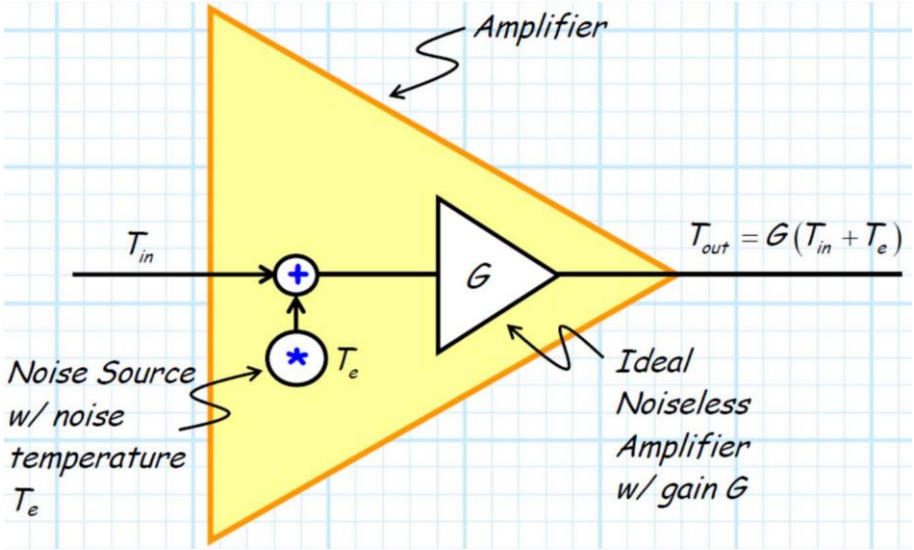
Any **white noise** source can be specified in terms of an **effective noise temperature**.

$$N_{ai} = (F-1)N_i$$

$$\Rightarrow \quad kT_e B = (F-1)kT_{in}B$$

$$\Rightarrow \quad T_e = (F-1)T_{in}$$

$$\Rightarrow \quad T_e = (F-1)\times 290\ \text{K}$$

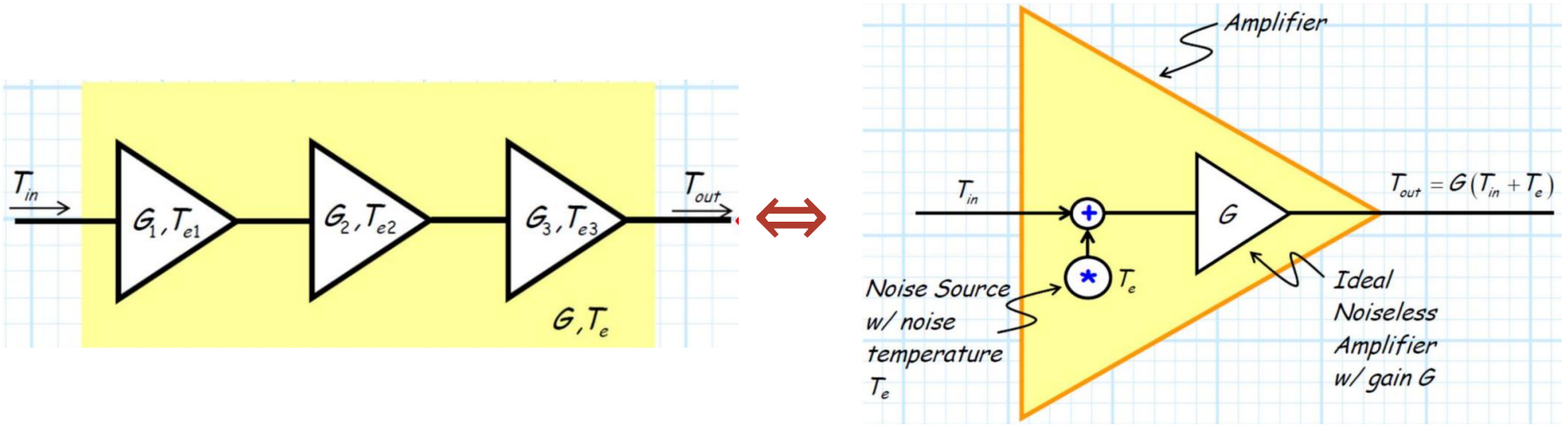


T_{in} is the reference temperature of the noise source, which is customarily taken as 290 K.

T_e is called **the effective (equivalent) input noise temperature** of a device (e.g., an amplifier).

- Noise temperature is an alternative to NF, but an equivalent characterization of the noisiness of a device.
- In general, applications involving very low noise devices seem to favor the effective temperature measure over the NF.

Noise Temperature (2): Equivalent Model



A cascade of stages is equivalent to one noiseless amplifier of gain G with an input noise source of temperature T_e

$$T_{out} = G(T_{in} + T_e)$$

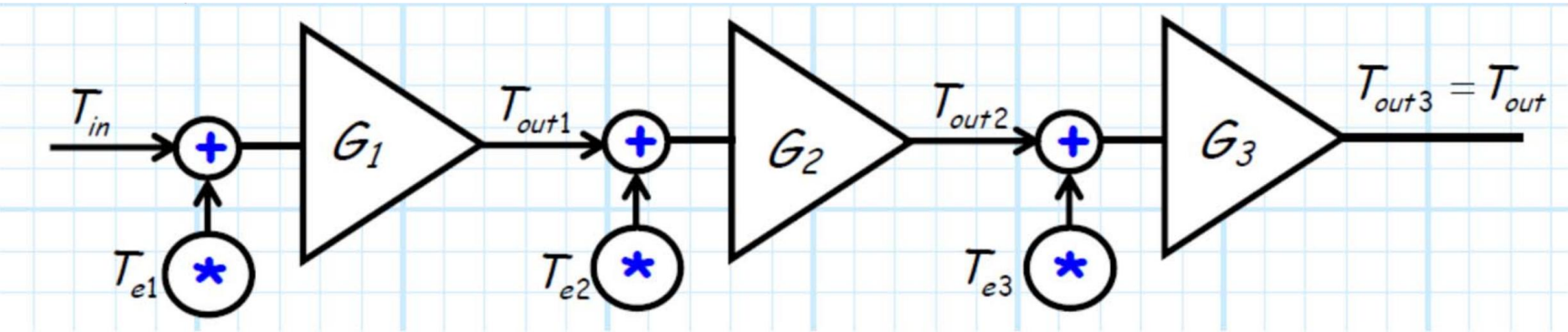
$$P_n^{out} = GkT_{in}B + GkT_eB = Gk(T_{in} + T_e)B$$

$$T_e = \frac{T_{out}}{G} - T_{in}$$

Overall gain of the cascade: $G = G_1 G_2 G_3$

Noise Temperature (3): Cascaded Stages

Each stage is replaced by a noiseless amplifier with its own input noise source:



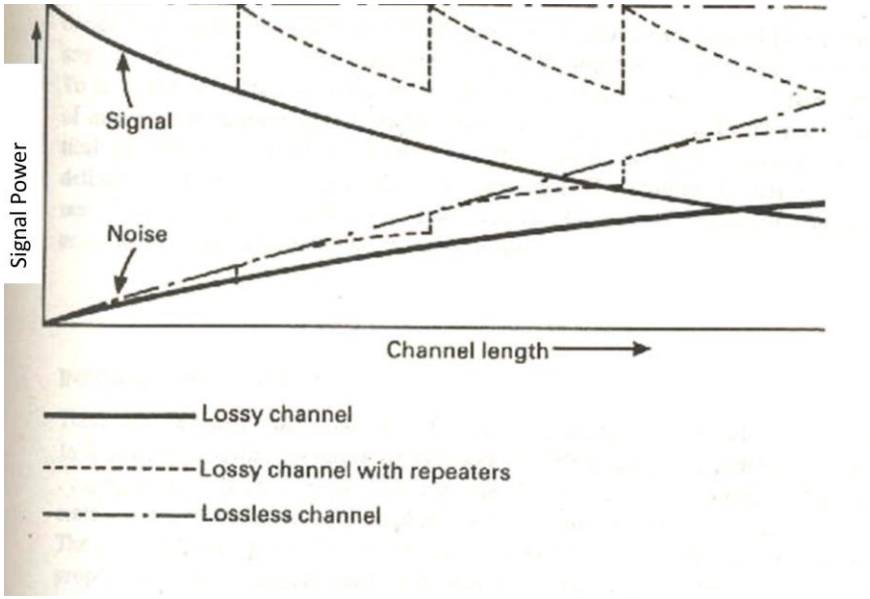
$$T_e = T_{e1} + \frac{T_{e2}}{G_1} + \frac{T_{e3}}{G_1 G_2} + \cdots + \frac{T_{eN}}{G_1 G_2 \cdots G_{N-1}}$$

- The noise temperature of each later stage is divided by the total gain that precedes it (compare with the cascaded noise-figure formula).

Encountering the Problem of Noise

Alternate options

1. **Increasing transmit power:** not feasible
2. **Increasing bandwidth:** very expensive
3. **Amplifier at the receiver side:** not useful, as it amplifies the noise as well
4. **Amplifiers along the line**
5. **Regenerative repeaters along the line**



Amplifiers along the line

dB, dBm, dBW

dBm and dBW are (logarithmic) **units to measure powers**.

$$P_T|_{\text{dBm}} = 10\log_{10}\left(\frac{P_T}{1\text{ mW}}\right)$$

$$P_T|_{\text{dBW}} = 10\log_{10}\left(\frac{P_T}{1\text{ W}}\right) = P_T|_{\text{dBm}} - 30\text{ dB}$$

dB is a logarithmic **ratio of powers**.

$$\frac{P_1}{P_2} \equiv 10\log_{10}\left(\frac{P_1}{P_2}\right)\text{dB} = P_1|_{\text{dBW}} - P_2|_{\text{dBW}} = P_1|_{\text{dBm}} - P_2|_{\text{dBm}}$$

dBW ± dB = dBW | **dBm ± dB = dBm** | **dBW – dBW = dB** | **dBm – dBm = dB**

A dB value can be added to, or subtracted from, an absolute level; the difference between two absolute levels is a ratio in dB.

dB Arithmetic: Common Pitfalls

Never add a bunch of quantities in dBW or dBm!

$$10\ \text{dBW} + 3\ \text{dBW} + 6\ \text{dBW} \quad \hat{=} \quad 10\ \text{W} \cdot 2\ \text{W} \cdot 4\ \text{W} = 80\ \text{W}^3 \quad \text{???}$$

Never ever multiply dBW with dB!

$$P_T \cdot G_T = 10\ \text{W} \cdot 10 = 100\ \text{W} \quad \hat{=} \quad 20\ \text{dBW}$$

$$\Leftrightarrow \quad P_T|_{\text{dBW}} + G_T|_{\text{dBi}} = 10\ \text{dBW} + 10\ \text{dBi} = 20\ \text{dBW}$$

Levels in dBW and gains in dB (or dBi) **can be added**.

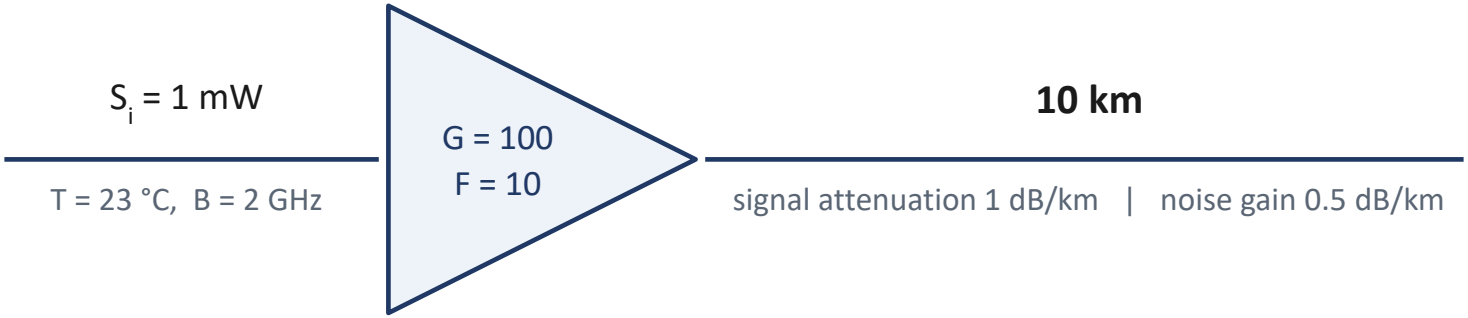
Example: Amplifier and Transmission Line

PROBLEM

Consider a system, as shown in the figure, that operates at T = 23 °C with a bandwidth of 2 GHz. The length of the transmission line is 10 km. Signal attenuation and noise gain along the transmission line are 1 dB/km and 0.5 dB/km respectively. The gain and the noise figure of the amplifier are 100 and 10 respectively. If the signal power at the input of the amplifier is 1 mW, calculate:

(i) noise power at the output of the amplifier

(ii) SNR at the end of the transmission line



Solution (i): Noise Power at the Amplifier Output

1 **Operating conditions**

$T = 23\ ^\circ C = 23 + 273 = 296\ K$ $\quad\quad$ $B = 2\ GHz = 2 \times 10^9\ Hz$

2 **Noise power at the amplifier input (thermal noise)**

$N_i = kTB = (1.38 \times 10^{-23}) \times 296 \times (2 \times 10^9)$

$N_i = 8.17 \times 10^{-12}\ W \approx$ **-80.9 dBm**

3 **Noise power at the amplifier output (G = 100, F = 10)**

$F = N_o / (G\, N_i) \quad \Rightarrow \quad N_o = F\, G\, N_i = 10 \times 100 \times 8.17 \times 10^{-12}\ W$

$N_o \approx 8.17 \times 10^{-9}\ W = 8.17\ nW \approx -50.9$ dBm

Check in dB: -80.9 dBm + 20 dB (gain) + 10 dB (noise figure) = -50.9 dBm

Uses $F = N_o / (G\, N_i)$ from the noise-figure slides, with $N_i = kTB$ at the operating temperature.

Solution (ii): SNR at the End of the Line

1 **Signal power at the amplifier output**

$S_o = G\,S_i = 100 \times 1\ \text{mW} = 100\ \text{mW} = \mathbf{20\ dBm}$

2 **After the 10 km transmission line**

Signal: 10 km × 1 dB/km = 10 dB loss ⇒ $S_{end} = 20 - 10 = \mathbf{10\ dBm}$ (10 mW)

Noise: 10 km × 0.5 dB/km = 5 dB gain ⇒ $N_{end} = -50.9 + 5 = \mathbf{-45.9\ dBm}$ (≈ 25.8 nW)

3 **SNR at the end of the line**

$SNR_{end} = S_{end} - N_{end} = 10\ \text{dBm} - (-45.9\ \text{dBm})$

SNR ≈ 55.9 dB (a ratio of about 3.9×10^5)

Point in the system	Signal	Noise	SNR
Amplifier input	0 dBm	−80.9 dBm	80.9 dB
Amplifier output	20 dBm	−50.9 dBm	70.9 dB
End of 10 km line	10 dBm	−45.9 dBm	**55.9 dB**

Key Takeaways

1. Digital messages use a **finite alphabet**. This gives immunity to noise and distortion and makes **regenerative repeaters** possible.

2. **Bandwidth B** and **signal power P_s** are the two primary communication resources, and they can be traded against each other.

3. **Shannon's equation** $C = B \log_2(1 + SNR)$ is the upper bound on the rate of reliable communication.

4. Thermal and shot noise are **white**. The maximum thermal noise power is **$P_n = kTB$**.

5. **Noise figure** $F = SNR_{in} / SNR_{out}$. Equivalently, the noise temperature is **$T_e = (F - 1) \times 290\ K$**.

6. A dB value is added to a level in dBm or dBW; two levels in dBm or dBW are **never added** together.